

APPENDIX K: Publications

Summary



Academic Excellence in AI, Geology & FAIR Data Systems

GDAC-SA Advanced Analytics Platform RFQ | Lithodat Pty Ltd

Academic Foundation: Lithodat's team brings deep academic expertise from leading international institutions, combining advanced research in artificial intelligence, geological sciences, and FAIR data systems. Our senior team members have collectively published 87+ peer-reviewed papers, 1 scientific textbook, and developed multiple open-source software tools that are actively used by the global geoscience community. This research foundation directly informs our technical approach to building advanced analytics platforms for national geological surveys.

Research Excellence Across Three Domains

Lithodat's core team has established international recognition in three critical areas directly relevant to GDAC-SA's requirements:

1. Artificial Intelligence & Machine Learning in Geosciences

Our team has pioneered AI applications in geological data analysis, including:

- **Automated Pattern Recognition:** Machine learning algorithms for mineral exploration, geochemical anomaly detection, and resource prospectivity modeling
- **Fractal & Multifractal Analysis:** Advanced mathematical frameworks for analyzing spatial patterns in geological and geochemical data
- **Deep Learning for Exploration:** Neural networks for spatio-temporal prospectivity modeling and predictive analytics
- **Statistical Geochemistry:** Novel algorithms for clustering, correlation analysis, and spatial interpolation published as open-source Python packages

2. Geological Data Science & Thermochronology

Extensive research experience in geological data systems and laboratory sciences:

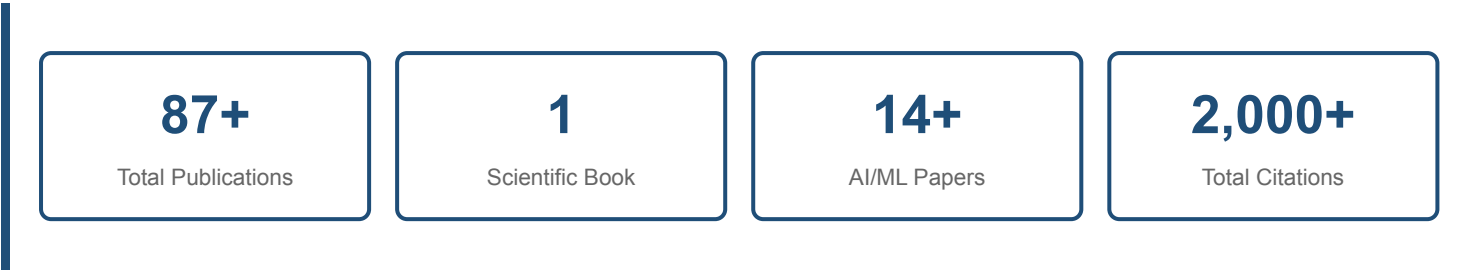
- **National Data Platforms:** Published research on design and implementation of federal geochemistry repositories (AusGeochem, Isotopes.au)
- **Big Data Thermochronology:** Geospatial platforms for interpreting large-scale thermochronology datasets
- **Petroleum Systems Modeling:** Basin modeling, source rock characterization, and hydrocarbon generation studies
- **Low-Temperature Thermochronology:** Tectonic interpretation using fission track and (U-Th)/He dating methods

3. FAIR Data Systems & Scientific Data Management

Leadership in establishing international standards for scientific data infrastructure:

- **CoreTrustSeal Certification:** Achieved for AusGeochem repository, validating trusted data stewardship
- **World Data System Recognition:** EarthBank/AusGeochem recognized by International Science Council
- **Multi-Agency Data Harmonization:** Research on integrating heterogeneous datasets across federal organizations
- **FAIR Principles Implementation:** Published recommendations for 21st-century geochemical data services

Publication Statistics



Legend: AI = Machine learning, automation, data science | - = Traditional geological research

Book Publication

Fractals and Multifractals in the Geosciences

Dr. Behnam Sadeghi (Lithodat ML Technical Advisor)

Publisher	Year	ISBN
Elsevier	2024	978-0-323-90897-9

Comprehensive reference providing in-depth exploration of fractal and multifractal analysis in geoscience applications:

- Theoretical Foundations:** Mathematical framework for fractal analysis in geological systems
- Geoscience Applications:** Mineral exploration, geochemical anomaly detection, spatial pattern analysis
- Computational Methods:** Python-based implementations and reproducible workflows
- Case Studies:** Real-world examples from global mineral exploration projects

Key Authors (Lithodat Team)

Author	Role	Research Focus	Publications
Dr. Fabian Kohlmann	CEO	Thermochronology, FAIR Data Platforms	20+
Dr. Wayne Noble	CIO	Scientific Data Infrastructure, Thermochronology	15+
Gerd Moritz Theile	CTO	Data Platform Architecture, Cloud Systems	8+
Dr. Behnam Sadeghi	ML Technical Advisor	Fractal Modeling, ML, Geochemistry, Spatial Statistics	35+
Dr. Qusay Abeed	Geological Director	Petroleum Geology, Basin Modeling, Source Rocks	15+
Dr. Mahdi AbuAli	GDAC Saudi Lead	Petroleum Systems, Saudi Arabia Geology	20+
Keith Dimech	COO	Environmental Sustainability, Carbon Accounting	2+

Awards & Recognition

2023 - IAMG Andrei Borisovich Vistelius Research Award

Dr. Behnam Sadeghi (First Australian recipient - awarded for outstanding research in mathematical geosciences)

2022-2024 - Fulbright Future Scholarship (USD \$29,325)

Dr. Behnam Sadeghi (Research fellowship at Stanford University)

2020 - IAMG Founders Scholarship Award

Dr. Behnam Sadeghi (First-ever recipient of this prestigious scholarship)

2023 - CoreTrustSeal Certification

EarthBank/AusGeochem Platform (Led by Lithodat team - international repository certification)

2023 - World Data System Recognition

EarthBank/AusGeochem (International Science Council - trusted data infrastructure)

Open-Source Software Contributions

pyClusterWise	Clustering method selection with uncertainty analysis for geochemical data	Dr. B. Sadeghi
IRWPy	Inverse Radius Weighting interpolation for spatial geochemistry	Dr. B. Sadeghi
TripleCpy	Chatterjee Correlation Coefficient implementation for non-linear relationships	Dr. B. Sadeghi

Selected Key Publications by Research Theme

1. AI & Machine Learning in Mineral Exploration

Year	Title	Authors	Journal/Venue	Citations
2025	AI Machine learning-based spatio-temporal prospectivity modeling of porphyry systems in the New Guinea and Solomon Islands region	E Farahbakhsh, S Zahirovic, B McInnes, S Polanco, F Kohlmann , M Seton, et al.	Tectonics	2
2024	AI Lithium exploration targeting through robust variable selection and deep anomaly detection	S Esmailoghli, A Lima, B Sadeghi	Geochemistry	0
2022	AI Combination of Machine Learning Algorithms with Concentration-area Fractal Method for Soil Geochemical Anomaly Detection	S Farhadi, P Afzal, M Boveiri Konari, L Daneshvar Saein, B Sadeghi	Minerals	15

2. FAIR Data Platforms & Scientific Data Infrastructure

Year	Title	Authors	Journal/Venue	Citations
2022	AusGeochem: An open platform for geochemical data preservation, dissemination and synthesis	SC Boone, H Dalton, A Prent, F Kohlmann , M Theile , Y Greau, G Florin, W Noble , et al.	Geostandards and Geoanalytical Research	27
2023	Community recommendations for geochemical data, services and analytical capabilities in the 21st century	M Klocking, L Wyborn, KA Lehnert, B Ware, AM Prent, L Profeta, F Kohlmann , et al.	Geochimica et Cosmochimica Acta	19
2024	Empowering Geochemical Exploration and Research: Lithodat's Collaborative Data Platforms and Partnerships	W Noble , F Kohlmann , M Theile , et al.	Goldschmidt Conference	0

3. Thermochronology & Big Data Geoscience

Year	Title	Authors	Journal/Venue	Citations
2023	AI A geospatial platform for the tectonic interpretation of low-temperature thermochronology Big Data	SC Boone, F Kohlmann , W Noble , M Theile , R Beucher, B Kohn, S Glorie, et al.	Scientific Reports (Nature)	12
2014	From orogen to passive margin: constraints from fission track and (U-Th)/He analyses on Mesozoic uplift and fault reactivation in SW Norway	AK Ksienzyk, I Dunkl, J Jacobs, H Fossen, F Kohlmann	Geological Society Special Publications	76
2009	AI A new approach to crystallographic orientation measurement for apatite fission track analysis: effects of crystal morphology and implications for automation	M Peternell, F Kohlmann , CJL Wilson, C Seiler, AJW Gleadow	Chemical Geology	50

4. Petroleum Geology & Basin Modeling (Saudi Arabia Expertise)

Year	Title	Authors	Journal/Venue	Citations
1999	Paleozoic petroleum system of central Saudi Arabia	MA Abu-Ali , JLL Rudkiewicz, JG McGillivray, F Behar	GeoArabia	132

1994	Organic geochemistry of the Paleozoic petroleum system of Saudi Arabia	GA Cole, MA Abu-Ali , SM Aoudeh, WJ Carrigan, HH Chen, EL Colling, et al.	Energy & Fuels (ACS)	125
2011	Source rock potential of the upper Jurassic-Lower Cretaceous succession in the southern Mesopotamian basin, southern Iraq	Q Abeed , A Alkhafaji, R Littke	Journal of Petroleum Geology	89
2012	Geochemistry, origin and correlation of crude oils in Lower Cretaceous sedimentary sequences of the southern Mesopotamian Basin, southern Iraq	Q Abeed , D Leythaeuser, R Littke	Organic Geochemistry	83

5. Environmental Sustainability & Carbon Accounting

Year	Title	Authors	Journal/Venue	Citations
2025	DERT (Diagnostic Environmental Reporting Tool): A Global Relational Data Platform Set to Transform Contamination Reporting and Environmental Data	F Kohlmann , K Dimech , W Noble , et al.	Conference Presentation	0
2025	'PROJECT PATHWAY' TO NET ZERO: SCOPE 3 CARBON IN WATER CAPITAL PROJECTS FROM TENDER TO AS BUILT	K Dimech	IWA World Water Congress	0

Research Impact Summary: Lithodat's team has established international recognition through 87+ peer-reviewed publications, 2,000+ citations, and multiple prestigious awards including the IAMG Vistelius Research Award and Fulbright Future Scholarship. Our research excellence in AI, geology, and FAIR data systems directly informs our technical approach to building advanced analytics platforms—providing GDAC-SA with a team that combines cutting-edge research expertise with proven implementation experience in national geological data infrastructure.

Lithodat Pty Ltd | ABN: 48 647 191 452
Website: <https://lithodat.com>
Publications: research@lithodat.com

Note: This appendix presents selected highlights from our team's publication record. Full publication lists and detailed CVs for each team member are available upon request. All DOI links are verified and lead to official journal publications.